

Question No 01:

Let $A = \{0, 1, 2, 3\}$ and define relations R , S , and T on A as follows:

$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$,

$S = \{(0, 0), (0, 2), (0, 3), (2, 3)\}$,

$T = \{(0, 1), (2, 3)\}$.

a. Is R reflexive? symmetric? transitive?

R is reflexive: There is a loop at each point of the directed graph. This means that each element of A is related to itself, so R is reflexive.

R is symmetric: In each case where there is an arrow going from one point of the graph to a second, there is an arrow going from the second point back to the first. This means that whenever one element of A is related by R to a second, then the second is related to the first. Hence R is symmetric.

R is not transitive: There is an arrow going from 1 to 0 and an arrow going from 0 to 3, but there is no arrow going from 1 to 3. This means that there are elements of A —0, 1, and 3—such that $1 R 0$ and $0 R 3$ but $1 \not R 3$. Hence R is not transitive.

b. Is S reflexive? symmetric? transitive?

S is not reflexive: There is no loop at 1, for example. Thus $(1, 1) \notin S$, and so S is not reflexive.

S is not symmetric: There is an arrow from 0 to 2 but not from 2 to 0. Hence $(0, 2) \in S$ but $(2, 0) \notin S$, and so S is not symmetric.

S is transitive: There are three cases for which there is an arrow going from one point of the graph to a second and from the second point to a third: Namely, there are arrows going from 0 to 2 and from 2 to 3; there are arrows going from 0 to 0 and from 0 to 2; and there are arrows going from 0 to 0 and from 0 to 3. In each case there is an arrow going from the first point to the third. (Note again that the “first,” “second,” and “third” points need not be distinct.) This means that whenever $(x, y) \in S$ and $(y, z) \in S$, then $(x, z) \in S$, for all $x, y, z \in \{0, 1, 2, 3\}$, and so S is transitive.

c. Is T reflexive? symmetric? Transitive?

T is not reflexive: There is no loop at 0, for example. Thus $(0, 0) \notin T$, so T is not reflexive.

T is not symmetric: There is an arrow from 0 to 1 but not from 1 to 0. Thus $(0, 1) \in T$ but $(1, 0) \notin T$, and so T is not symmetric.

T is transitive by default because it is not not transitive! T is transitive: The transitivity condition is vacuously true for T . To see this, observe that the transitivity condition says that For all $x, y, z \in A$, if $(x, y) \in T$ and $(y, z) \in T$ then $(x, z) \in T$.

Question No 02:

Let $A = \{0, 1, 2, 3\}$ and consider the relation R defined on A as follows:

$R = \{(0, 1), (1, 2), (2, 3)\}$.

Find the transitive closure of R .

$(0, 2)$ AND $(1, 3)$

Question No 03 In 1–8 a number of relations are defined on the set $A = \{0, 1, 2, 3\}$. For each relation:

a. Draw the directed graph.

b. Determine whether the relation is reflexive.

c. Determine whether the relation is symmetric.

d. Determine whether the relation is transitive.

Give a counterexample in each case in which the relation does not satisfy one of the properties.

1. $R_1 = \{(0, 0), (0, 1), (0, 3), (1, 1), (1, 0), (2, 3), (3, 3)\}$

NOT REFLEXIVE

Not symmetric

Not transitive

2. $R_2 = \{(0, 0), (0, 1), (1, 1), (1, 2), (2, 2), (2, 3)\}$

3. $R_3 = \{(2, 3), (3, 2)\}$

4. $R_4 = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$

5. $R_5 = \{(0, 0), (0, 1), (0, 2), (1, 2)\}$

6. $R_6 = \{(0, 1), (0, 2)\}$

7. $R_7 = \{(0, 3), (2, 3)\}$

8. $R_8 = \{(0, 0), (1, 1)\}$

Question No 04: determine whether the given relation is reflexive, symmetric, transitive, or none of these. Justify your answer

1. **R is the “greater than or equal to” relation on the set of real numbers: For all $x, y \in \mathbb{R}$, $xRy \Leftrightarrow x \geq y$.**

1. Reflexivity:

For every real number x , x is always greater than or equal to itself. Therefore, the relation R is reflexive.

2. Symmetry:

For every pair (x, y) in R , where $x \geq y$, it does not necessarily imply that $y \geq x$. Hence, the relation R is not symmetric.

3. Transitivity:

For any three real numbers x , y , and z , if $x \geq y$ and $y \geq z$, then it follows that $x \geq z$. This satisfies the transitive property. Therefore, the relation R is transitive.

2. **C is the circle relation on the set of real numbers: For all $x, y \in \mathbb{R}$, $xCy \Leftrightarrow x^2 + y^2 = 1$.**

(a) C is not reflexive. To be reflexive, for every real number, xCx must hold true, meaning $x^2 + x^2 = 1$. This is false. For example, let $x=0$. In this case, $x^2 + x^2 = 0$, which does not equal 1. Therefore, C is not reflexive.

(b) C is symmetric. If xCy then yCx , for all real numbers x and y . If we see the definition of C , this means that if $x^2 + y^2 = 1$, then $y^2 + x^2 = 1$. This is true due to the commutative property of addition ($x^2 + y^2 = y^2 + x^2$ for all real numbers x and y). Thus, C is symmetric.

(c) C is not transitive. To be transitive, if xCy and yCz , then xCz must hold true for all real numbers x , y , and z . This means that if $x^2 + y^2 = 1$ and $y^2 + z^2 = 1$, then $x^2 + z^2$ must equal 1. This is not always true. Let's take an example $(x, y, z) = (1, 0, -1)$. Then $x^2 + y^2 = 1$, $y^2 + z^2 = 1$, but $x^2 + z^2 = 2$, not 1. Thus, C is not transitive.

3. **D is the relation defined on $\mathbb{R}$ as follows: For all $x, y \in \mathbb{R}$, $xDy \Leftrightarrow x - y \geq 0$.**

1. Reflexivity:

For every real number x , $x*x = x^2$, which is always greater than or equal to 0. Therefore, the relation D is reflexive.

2. Symmetry:

For every pair (x, y) in D , where $x*y \geq 0$, it is also true that $y*x \geq 0$. Therefore, the relation D is symmetric.

3. Transitivity:

For D to be transitive means that for all real numbers x, y , and z , if $x D y$ and $y D z$ then $x D z$. By definition of D , this means that for all real numbers x, y , and z , if $x y \geq 0$ and $y z \geq 0$ then $x z \geq 0$. But this is false: there exist real numbers x, y , and z such that $xy \geq 0$ and $y z \geq 0$ but $x z < 0$. As a counterexample, let $x = 1, y = 0$, and $z = -1$. Then $x D y$ and $y D z$ because $1 \cdot 0 \geq 0$ and $0 \cdot (-1) \geq 0$. But $x D z$ because $1 \cdot (-1)$ is not greater than equal to 0. Therefore, the relation D is not transitive

4. E is the congruence modulo 2 relation on Z : For all $m, n \in Z$, $m E n \Leftrightarrow 2 \mid (m - n)$.

1. Reflexivity:

For every integer m , $m - m = 0$, and 2 divides 0. Therefore, the relation E is reflexive.

2. Symmetry:

For every pair (m, n) in E , where 2 divides $(m - n)$, it is also true that 2 divides $(n - m)$. Therefore, the relation E is symmetric.

3. Transitivity:

For any three integers m, n , and k , if 2 divides $(m - n)$ and 2 divides $(n - k)$, then it follows that 2 divides $(m - k)$. This satisfies the transitive property. Therefore, the relation E is transitive.

5. F is the congruence modulo 5 relation on Z : For all $m, n \in Z$, $m F n \Leftrightarrow 5 \mid (m - n)$.

1. Reflexivity:

For every integer m , $m - m = 0$, and 5 divides 0. Therefore, the relation F is reflexive.

2. Symmetry:

For every pair (m, n) in F , where 5 divides $(m - n)$, it is also true that 5 divides $(n - m)$. Therefore, the relation F is symmetric.

3. Transitivity:

For any three integers m, n , and k , if 5 divides $(m - n)$ and 5 divides $(n - k)$, then it follows that 5 divides $(m - k)$. This satisfies the transitive property. Therefore, the relation F is transitive.

Since the relation F is reflexive, symmetric, and transitive, we can conclude that F is an equivalence relation.

6. O is the relation defined on Z as follows: For all $m, n \in Z$, $m O n \Leftrightarrow m - n$ is odd.

1. Reflexivity:

For every integer m , $m - m = 0$, which is an even number. Therefore, the relation O is not reflexive, since an even number subtracted from itself is always even, not odd.

2. Symmetry:

for all $x, y \in Z$, if $x O y$ then $y O x$
 $x O y \Leftrightarrow x - y = 2k + 1$

$-x+y = -2k - 1$ (Multiply both sides by -1)
 $y - x = 2\{-k\} - 1$ (Commutative Law)
 $y - x = \text{odd}$ (note that $-k \in \mathbb{Z}$ hence $y - x$ is odd)
 yOx (by Def of O)

3. Transitivity:

For any three integers m , n , and k , if $m - n$ is odd and $n - k$ is odd, it does not necessarily imply that $m - k$ is odd. The sum or difference of two odd numbers can result in an even number. Therefore, the relation O is not transitive.

7. D is the “divides” relation on $\mathbb{Z}^+$: For all positive integers m and n , $mDn \Leftrightarrow m \mid n$.

1. Reflexivity:

For every positive integer m , m divides itself, since m multiplied by 1 equals m . Therefore, the relation D is reflexive.

2. Symmetry:

For every pair (m, n) in D , where m divides n , it is not necessarily true that n divides m . The divisibility relation is not symmetric. For example, if 3 divides 6 , it does not imply that 6 divides 3 . Therefore, the relation D is not symmetric.

3. Transitivity:

Transitive: for all x, y and $z \in \mathbb{Z}$ if $(xOy \text{ and } yOz)$ then (xOz)

Counter Example: take $x = 1, y = 2, z = 1$

$xOy = -1 \# \text{odd}$

$yOz = 1 \# \text{odd}$

but $xOz = 0 \# \text{even}$, hence the statement is false, hence not transitive

8. A is the “absolute value” relation on $\mathbb{R}$: For all real numbers x and y , $xAy \Leftrightarrow |x| = |y|$.

1. Reflexivity:

For every real number x , $|x| = |x|$. Therefore, the relation A is reflexive.

2. Symmetry:

For every pair (x, y) in A , where $|x| = |y|$, it is also true that $|y| = |x|$. Therefore, the relation A is symmetric.

3. Transitivity:

For any three real numbers x , y , and z , if $|x| = |y|$ and $|y| = |z|$, then it follows that $|x| = |z|$. This satisfies the transitive property. Therefore, the relation A is transitive.

Question No 05:

Define a relation Q on $\mathbb{R}$ as follows: For all real numbers x and y , $xQy \Leftrightarrow x - y$ is rational.

1. Reflexivity:

For every real number x , $x - x = 0$, which is a rational number. Therefore, the relation Q is reflexive.

Reflexive: for all $x \in \mathbb{Z}$ xAx

true since $x - x = 0$ (note that 0 is rational $= 0/1$)

2. Symmetry:

For every pair (x, y) in Q , where $x - y$ is rational, it is also true that $y - x = -(x - y)$ is rational.

Therefore, the relation Q is symmetric.

for all $x, y \in \mathbb{Z}$ if xQy then yQx

$xQy \Leftrightarrow x - y = ab \text{ } b \neq 0$

$-x+y = -ab$ (multiplied by -1)
 $y - x = -ab$ (commutative, $b \neq 0$ hence rational)
 yQx by definition of Q

3. Transitivity:

For any three real numbers x , y , and z , if $x - y$ is rational and $y - z$ is rational, then it follows that $x - z = (x - y) + (y - z)$ is also rational. This satisfies the transitive property. Therefore, the relation Q is transitive.

Transitive: for all x, y , and $z \in \mathbb{Z}$ if $(xQy \text{ and } yQz)$ then xQz

$xQy \Leftrightarrow x-y = ab, b \neq 0$

$yQz \Leftrightarrow y-z = cd, d \neq 0$

add (1) and (2)

$x-z = ad+cb, bd \neq 0$ ($x-z$ is rational)

xQz (By definition of Q) [that's what we needed to show]

Question No 06:

Define a relation I on $\mathbb{R}$ as follows: For all real numbers x and y , $xIy \Leftrightarrow x - y$ is irrational.

1. Not Reflexivity:

or all $x \in \mathbb{Z}$ xIx

false since $x-x = 0$ (note that 0 is rational = 0/1)

2. Symmetry:

Symmetric: for all $x, y \in \mathbb{Z}$ if xIy then yIx

$xIy \Leftrightarrow x-y = \text{irrational number } X$

$-x+y = -X$ (multiplied by -1)

$y - x = -X$ (commutative law, adding -Sign won't change the fact that X is irrational)

hence yIx by definition of I .

3. Not Transitivity:

For any three real numbers x , y , and z , if $x - y$ is irrational and $y - z$ is irrational, it does not necessarily imply that $x - z$ is irrational. The sum or difference of two irrational numbers can result in a rational number. Therefore, the relation I is not transitive. Transitive: for all x, y , and $z \in \mathbb{Z}$ if $(xIy \text{ and } yIz)$ then xIz

let $x = 2\sqrt{2}$, $y = 1$, $z = 2\sqrt{2}$

$xIy \Leftrightarrow x-y = 2\sqrt{2} - 1$, irrational

$yIz \Leftrightarrow y-z = 1 - 2\sqrt{2}$, irrational

but $xIz \Leftrightarrow 2\sqrt{2} - 2\sqrt{2} = \text{rational} \{ \text{hence false} \}$

Question No 07:

Let $X = \{a, b, c\}$ and $P(X)$ be the power set of X (the set of all subsets of X). A relation E is defined on $P(X)$ as follows: For all $A, B \in P(X)$, $A E B \Leftrightarrow$ the number of elements in A equals the number of elements in B .

1. Reflexivity:

$-E$ is reflexive $\Leftrightarrow$ for all sub sets A of X , $A E A$. By definition of E , this means that for all subsets A of X , A has the same number of elements as A . But this is true.

2. Symmetry:

E is symmetric $\Leftrightarrow$ for all subsets A and B of X , if $A E B$ then $B E A$. By definition of E , this means that if A has the same number of elements as B , then B has the same number of elements as A . But this is true.

3. Transitivity:

E is transitive $\Leftrightarrow$ for all subsets A, B, and C of X, if $A \in B$ and $B \in C$, then $A \in C$. By definition of E, this means that for all subsets, A, B, and C of X, if A has the same number of elements as B and B has the number of elements as C, then A has the same number of elements as C. But this is true.

Question No 08:

Let $X = \{a, b, c\}$ and $P(X)$ be the power set of X. A relation L is defined on $P(X)$ as follows: For all $A, B \in P(X)$, $A L B \Leftrightarrow$ the number of elements in A is less than the number of elements in B.

1. Not Reflexivity:

for all $B \in P(X)$ $B L B$,

the statement is false since it invalidates the relation L since a set has the same number of elements of the set itself.

3. Not Symmetry:

Symmetric: for all $A, B \in P(X)$ if $A L B$ then $B L A$

assume that $A L B$ is true which means set A has fewer elements than set B which means that $B L A$ is false since set B has more elements than A.

3. Transitivity:

Transitive: for all A, B , and $C \in P(X)$ if $A L B$ AND $B L C$ then $A L C$.

$A L B \Leftrightarrow A B$

Question No 09:

Let $X = \{a, b, c\}$ and $P(X)$ be the power set of X. A relation N is defined on $P(X)$ as follows: For all $A, B \in P(X)$, $A N B \Leftrightarrow$ the number of elements in A is not equal to the number of elements in B.

1. Not Reflexivity:

Reflexive: for every $A \in P(X)$ $A N A$ it invalidates the relation since the number of elements in A is the same number of elements in A itself.

2. Symmetry:

Symmetric: for every $A, B \in P(X)$ if $A N B$ THEN $B N A$ let $A N B$ be true, which means A doesn't have equal elements as B, which also implies that B doesn't have equal elements as A, then by definition of N $B N A$ (HENCE TRUE).

3. Not Transitivity:

If $A N B$ and $B N C$ then $A N C$ for all $A, B, C \in P(X)$

let A be a set with 8 elements, B with 5, C with 8.

which implies that

$A N B$ and $B N C$

but $A N C$ is false since A and C have 8 elements each.

which implies that the relation is not transitive.

Question No 10:

Let X be a nonempty set and $P(X)$ the power set of X. Define the "subset" relation S on $P(X)$ as follows:

For all $A, B \in P(X)$, $A S B \Leftrightarrow A \subseteq B$

1. Reflexivity:

For any subset A of X, A is always a subset of itself. Therefore, $A S A$ for all subsets A of X. Hence, the relation S is reflexive.

2. Antisymmetry:

For any subsets A and B of X , if $A \subseteq B$ and $B \subseteq A$, it means that $A \subseteq B$ and $B \subseteq A$. From this, we can conclude that $A = B$. Therefore, if $A \subseteq B$ and $B \subseteq A$, then $A = B$. Hence, the relation S is antisymmetric.

3. Transitivity:

For any subsets A , B , and C of X , if $A \subseteq B$ and $B \subseteq C$, it means that $A \subseteq B$ and $B \subseteq C$. From this, we can conclude that $A \subseteq C$. Therefore, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. Hence, the relation S is transitive.